WORKED SOLUTION

STEP II Specimen paper, Question 14

Mechanics • Rotation of Rigid Bodies

STEP II Specimen paper, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q14

The key idea

Both the disc and the particle turn about the same fixed vertical axis, so the natural tool is the rotational equation of motion, torque = moment of inertia × angular acceleration, applied to each body separately. The friction between them is an internal pair of forces: it speeds one up exactly as much (in angular momentum) as it slows the other down. That gives the conserved combination at once, and subtracting the two equations gives a first-order equation for the *relative* angular speed, which decays exponentially.

Solution

The model

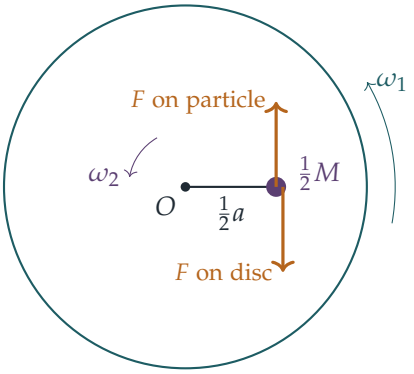
The disc is a uniform lamina turning about a smooth fixed axis through its centre O, so its moment of inertia about the axis is

$$I_1 = \tfrac{1}{2}Ma^2.$$

The particle, of mass $\frac{1}{2}M$, is held by a light rigid wire at a fixed distance $\frac{1}{2}a$ from the axis, so it moves on a circle of radius $\frac{1}{2}a$ and its moment of inertia about the axis is

$$I_2 = \tfrac{1}{2}M\left(\tfrac{1}{2}a\right)^2 = \tfrac{1}{8}Ma^2.$$

Initially $\omega_1 = \Omega$ and $\omega_2 = 0$, so the disc is moving faster than the particle at the point of contact. Friction therefore acts on the particle in its direction of motion (tangentially, at radius $\frac{1}{2}a$), and by Newton's third law an equal and opposite force acts on the disc at the same point.



(View from above; $F = Mak(\omega_1 - \omega_2)$. The tension in the wire is radial, so it has no moment about the axis.)

Equations of motion

The friction force $F = Mak(\omega_1 - \omega_2)$ acts at distance $\frac{1}{2}a$ from the axis, perpendicular to the radius, so its moment about the axis is

$$F \cdot \tfrac{1}{2}a = \tfrac{1}{2}Ma^2k(\omega_1 - \omega_2).$$

The weight of the particle and the normal reactions are vertical, so they have no moment about the vertical axis, and the axis itself is smooth. For the disc, friction opposes the rotation:

$$\tfrac{1}{2}Ma^2\,\frac{\mathrm{d}\omega_1}{\mathrm{d}t} = -\tfrac{1}{2}Ma^2k(\omega_1 - \omega_2) \quad\Longrightarrow\quad \frac{\mathrm{d}\omega_1}{\mathrm{d}t} = -k(\omega_1 - \omega_2).$$

For the particle, friction drives the rotation:

$$\tfrac{1}{8}Ma^2\,\frac{\mathrm{d}\omega_2}{\mathrm{d}t} = \tfrac{1}{2}Ma^2k(\omega_1 - \omega_2) \quad\Longrightarrow\quad \frac{\mathrm{d}\omega_2}{\mathrm{d}t} = 4k(\omega_1 - \omega_2).$$

(For the particle you can equally use Newton's second law tangentially: $\frac{1}{2}M \cdot \frac{1}{2}a\,\dot{\omega}_2 = F$, which is the same equation.)

The conserved combination

Multiplying the first equation by 4 and adding the second,

$$4\frac{\mathrm{d}\omega_1}{\mathrm{d}t} + \frac{\mathrm{d}\omega_2}{\mathrm{d}t} = -4k(\omega_1 - \omega_2) + 4k(\omega_1 - \omega_2) = 0,$$

so $4\omega_1 + \omega_2$ is constant. At $t = 0$ it equals $4\Omega + 0$, hence

$$4\omega_1 + \omega_2 = 4\Omega,$$

as required. This is conservation of angular momentum about the axis: $I_1\omega_1 + I_2\omega_2 = \frac{1}{8}Ma^2(4\omega_1 + \omega_2)$, and the only external forces (the reaction at the axis, gravity and the vertical reactions) have no moment about it.

Solving for ω_1

> **Ask yourself** We have two coupled equations. Which single quantity satisfies an equation on its own?

Both rates depend only on the difference $u = \omega_1 - \omega_2$, so look at that. Subtracting the equations,

$$\frac{\mathrm{d}u}{\mathrm{d}t} = \frac{\mathrm{d}\omega_1}{\mathrm{d}t} - \frac{\mathrm{d}\omega_2}{\mathrm{d}t} = -ku - 4ku = -5ku,$$

with $u(0) = \Omega - 0 = \Omega$. Separating variables (or recognising exponential decay),

$$\omega_1 - \omega_2 = u = \Omega\,\mathrm{e}^{-5kt}.$$

Adding this to $4\omega_1 + \omega_2 = 4\Omega$ eliminates ω_2:

$$5\omega_1 = 4\Omega + \Omega\,\mathrm{e}^{-5kt}, \qquad \text{so} \qquad \omega_1 = \frac{\Omega}{5}\left(4 + \mathrm{e}^{-5kt}\right),$$

as required. It follows that $\omega_2 = 4\Omega - 4\omega_1 = \frac{4}{5}\Omega\left(1 - \mathrm{e}^{-5kt}\right)$.

Alternatively, substitute $\omega_2 = 4\Omega - 4\omega_1$ into the disc's equation to get $\dot{\omega}_1 = -5k\omega_1 + 4k\Omega$, a linear equation whose solution with $\omega_1(0) = \Omega$ is the same.

Checks

- At $t = 0$: $\omega_1 = \frac{1}{5}\Omega(4+1) = \Omega$ and $\omega_2 = 0$. Correct.
- As $t \to \infty$: $\omega_1, \omega_2 \to \frac{4}{5}\Omega$, a common speed, as friction must eventually produce. The final kinetic energy is $\frac{1}{2} \cdot \frac{5}{8}Ma^2 \cdot \frac{16}{25}\Omega^2 = \frac{1}{5}Ma^2\Omega^2$, against $\frac{1}{4}Ma^2\Omega^2$ initially, so a fifth of the energy is lost to friction, as expected in a "perfectly inelastic" coupling.
- Numerically, with $M = a = k = \Omega = 1$, stepping the two equations of motion to $t = 1$ gives $\omega_1 = 0.80135$ and $\omega_2 = 0.79461$, matching $\frac{1}{5}(4 + \mathrm{e}^{-5})$ and $\frac{4}{5}(1 - \mathrm{e}^{-5})$, with $4\omega_1 + \omega_2$ staying at 4 throughout.

Common slips

- Taking the particle's moment of inertia as $\frac{1}{2}Ma^2$ (as if it were at the rim) instead of $\frac{1}{2}M(\frac{1}{2}a)^2$.
- Using the friction force itself, rather than its moment $F \cdot \frac{1}{2}a$, in the rotational equations.
- Giving friction the same sign in both equations: it slows the disc and speeds up the particle.

Beyond STEP

The friction here is an internal force, and the conservation of $4\omega_1 + \omega_2$ is the rotational version of momentum conservation in a collision; the exponential approach to a common speed is the same mathematics as two bodies coupled by viscous drag. More generally, a linear system $\dot{\mathbf{x}} = A\mathbf{x}$ decouples when written in terms of the eigenvectors of A: here they are the combinations $4\omega_1 + \omega_2$ (eigenvalue 0) and $\omega_1 - \omega_2$ (eigenvalue $-5k$). Kleppner and Kolenkow, *An Introduction to Mechanics*, develops rigid-body rotation and angular momentum carefully, and Boyce and DiPrima, *Elementary Differential Equations*, treats linear systems by eigenvectors.

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